TEXORA · LECTURE SHEET

Recent Advancements in Wet Processing Machineries and Technologies

The direction of forty years of machine development can be written in one ratio: liquor ratio down, control up. This sheet surveys the technologies that define a modern dyehouse — and quantifies what they save.

WP-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 11 · SPECIALIZATION — WET PROCESSING

1. The drivers: water, energy, time, right-first-time

Wet processing is the most resource-hungry part of textile manufacturing: water as solvent and wash medium, steam as heat, chemicals as auxiliaries, and time on expensive machines. Every advancement in this sheet attacks one of four costs — **water, energy, chemicals, or reprocessing** — and the honest way to present each is the way this course has worked throughout: with numbers, labelled as typical where they vary between installations.

2. Low-liquor dyeing machinery

Modern jets run at LR 1:4–1:6 where their predecessors needed 1:10–1:15, through redesigned liquor circuits, fabric transport that works with less bath, and precise level control. **Airflow machines** replace most of the transport liquor with an air stream: the rope moves in air while a small liquor volume is sprayed and exchanged continuously, reaching LR ≈ 1:3–1:5.

WORKED EXAMPLE 1 — WHAT A LIQUOR-RATIO STEP IS WORTH

A plant dyes 1,000 kg of fabric per day in batch machines.

Conventional jet at LR 1:10: dye bath alone = 1,000 × 10 = **10,000 L/day**.

Airflow at LR 1:4: bath = 1,000 × 4 = **4,000 L/day** — a saving of **6,000 L/day** in the dye bath alone.

Because salt and auxiliaries are dosed in g/L, their consumption falls in the same proportion (at 60 g/L, salt falls from 600 kg/day to 240 kg/day for the dye bath), and the steam to heat the bath falls

> roughly with its volume. Rinsing adds further savings of the same character. Low LR is simultaneously a water, chemical, energy and effluent technology.

3. Waterless and near-waterless colouration

- **Supercritical CO_2 dyeing:** carbon dioxide above its critical point (31 °C, 74 bar) behaves as a solvent for disperse dyes; polyester is dyed with no water at all, excess dye is recovered, and the CO_2 is recycled in the sealed system. Commercialised for polyester; the benchmark of "waterless" claims.
- **Dope (solution) dyeing:** pigment is added to the polymer melt before extrusion, so the fibre is born coloured — no dyehouse water for those shades, with outstanding fastness; limited to spun-dyed product programmes planned at the fibre stage.
- **Foam application:** chemicals or dyes applied as a foam of very low wet pick-up (Topic WP-4203), collapsing on the fabric — large water and drying-energy savings in finishing.
- **Digital printing:** colour placed only where the design needs it, with minimal wash-off (Topic 8).

4. Automation, dosing and control

- **Automatic dispensing:** dyes and chemicals weighed/metered by machine from recipe data — eliminating the weighing errors behind many shade failures, and logging every addition for traceability.
- **Centralised machine controllers and plant networks:** temperature profiles, dosing curves and cycle data are recipe-driven; production monitoring flags a drifting machine before it produces seconds.
- **Inline measurement:** spectrophotometric sampling of dyebaths (exhaustion curves followed in real time), moisture and temperature sensing on stenters, pH/conductivity control of wash waters.
- **Right-first-time systems:** the combination of calibrated prediction (Topics 1 and 3), dispensing accuracy and machine control raises RFT, and every point of RFT is worth more than any single utility saving, because a re-dye doubles all inputs.

5. Denim and garment technologies

- **Ozone washing:** gas-phase oxidation lightens indigo and clears back-staining with little water and no hypochlorite (Topic 10).
- **Laser finishing:** digitally programmed surface ablation replaces hand sanding and PP spray for many effects (Topic 10).

- **Enzyme systems:** engineered cellulases with anti-back-staining performance reduce stone use and machine wear.
- **Water recycling in wash plants:** screening, and membrane/biological treatment returning part of the wash water to process.

6. Energy technologies

TECHNOLOGY	SAVING MECHANISM
Heat recovery from hot effluent	Plate exchangers pre-heat incoming process water with outgoing hot drain water — the classic first payback in any dyehouse audit
Low-LR machines	Less water to heat per kilogram (Worked Example 1)
Stenter exhaust and moisture control	Exit-moisture feedback stops over-drying; exhaust heat recovery pre-heats air/water
Insulation and steam-trap management	Un glamorous, continuous savings on every steam line
Combined heat and power / solar pre-heating	Site-level supply economics for steam-heavy plants

All savings figures in this sheet are typical orders of magnitude for orientation; any investment case must be built on the plant's own metered data.

7. Glossary — key terms

- **Supercritical CO_2:** carbon dioxide held above its critical temperature and pressure, used as a dyeing solvent.
- **Dope dyeing:** colouring fibre at the polymer stage, before extrusion.
- **Right-first-time (RFT):** the percentage of lots meeting specification without correction.
- **Inline measurement:** sensing performed continuously within the process rather than by sampling afterwards.

8. Key points

- Judge every "advancement" by its quantified effect on water, energy, chemicals or reprocessing — as in Worked Example 1.
- Low LR, automation and RFT multiply each other: less bath, dosed exactly, dyed once.

- Waterless technologies ($scCO_2$, dope dyeing) are real but bounded by fibre and product programme; they complement rather than abolish the dyehouse.

9. Summary

The modern dyehouse differs from its predecessor less in chemistry than in ratios and control: liquor ratios halved, additions metered to the gram, exhaustion watched as it happens. The engineer's task is to evaluate each technology on metered numbers — this course's recurring discipline — and to build the investment case as carefully as a recipe.

10. Practice

1. A plant dyes 2,500 kg/day. Compare daily dye-bath volume and dye-bath salt use (at 50 g/L) between LR 1:12 machines and LR 1:5 machines.
2. Why does raising right-first-time from 85% to 95% save more than the 10 points suggest at first glance?
3. Give one limitation each for supercritical CO_2 dyeing and dope dyeing as general replacements for exhaust dyeing.

Answer key

1. LR 1:12: bath = 2,500 × 12 = 30,000 L/day; salt = 50 × 30,000/1000 = 1,500 kg/day. LR 1:5: bath = 12,500 L/day; salt = 625 kg/day. Saving: 17,500 L of water and 875 kg of salt per day, in the dye bath alone.
2. Each failed lot consumes a second full set of water, chemicals, steam and machine time, plus correction labour and delay. Moving 10% of lots out of the reprocess stream removes roughly 10% of total inputs, not 10% of one input — and releases machine capacity (Topic 7 arithmetic) for new production instead of repetition.
3. $scCO_2$: commercially proven essentially for disperse dyeing of polyester (and pressure-equipment capital cost); natural fibres lack a general route. Dope dyeing: shades must be committed at fibre production — it cannot respond to fashion shade changes or small lots, and it colours only synthetic filament/staple programmes planned for it.